

Question 1 (10 points): Who are the *stakeholders* here? For each stakeholder, briefly explain the most important *values* they may have that are at stake in this discussion.

- Direct Stakeholders (and associated values)
- Indirect Stakeholders (and associated values)

Direct Stakeholders:

Shareholders: The most important value for these shareholders is that the company succeeds, meaning that it continues to accrue profit and make their initial investments worthwhile.

Ourselves, the developers: The developers care about the reputation of their product; they want to create something that won't receive significant backlash by being deemed unethical, while simultaneously supporting their values ("a place that is as open as possible to its users' ability to 'post what they want'"). Additionally, the developers want to make good on the investments of their shareholders as well as the values of other sources of income (such as advertisers or content creators).

Content creators: Content creators will likely value the freedom to post whatever they like, as well as the ability to generate a form of revenue from their posts. Additionally, with the recent TikTok data scandal, it is likely that content creators would want their information to be stored securely on the app, as well as for the reputation of the app to be positive.

Site users: The average user's values likely revolve around a desire for entertainment and, similar to the developers' values, to stay away from disinformation and be informed about the world around them. Drawing from prior experience, users would also value *not* becoming addicted or mentally afflicted by the app.

Advertisers: Advertisers are direct stakeholders who invest money in the company in exchange for the advertisement of the specified product. They value achieving the maximum number of possible views (and clicks) for their product on the app, regardless of

the means. Advertisers would likely benefit from the removal of harmful disinformation, but this value isn't as important to them as overall "air time".

Indirect Stakeholders:

Journalists: Journalists value the ability to have their information re-posted and not construed by disinformation on the app. For example, the news a journalist puts out may be influenced by how many reposts it receives. In this regard, if users on the platform are already influenced by contradicting news being posted, they will be less likely to repost the journalist's reports. Such an event could easily affect the journalist, causing them to conform to the contradicting news on the app to increase their own influence/number of reposts.

Political Organizations: Arguably a direct stakeholder, but because they don't necessarily need to be involved in the app I put them as indirect. Political organizations such as the Democratic or Republican party value their influence above all else; influence wins votes, and votes win elections. In this regard, social media apps present themselves as a powerful tool that can both help political organizations or harm them, regardless of their direct presence on the app. Additionally, while these organizations likely value the removal of disinformation that could harm *themselves*, some may be in support of disinformation that harms their competition.

Large Businesses: According to Forbes, 87% of executives believe that disinformation is a significant risk to the reputation of modern-day businesses. In this regard, businesses would likely value the protection and upkeep of their reputation on the app, wanting to mitigate the spread and (more importantly) impact of harmful disinformation aimed at their company (regardless of whether or not they have an account on the site).

Non-users: People who don't use the app will likely be indirectly affected by reposts to sites/parts of the internet that they *do* interact with. Users who value staying away from social media or the app in particular may be negatively influenced by this result, however it could also help with "spreading the word" about the app to potential new users who wouldn't have heard about it otherwise.

Question 2 (10 points): A crucial part of VSD analysis is technical inquiry. That involves a careful investigation of what technical features of a technology are possible, and how they might be implemented in a way that is consistent with important values. Based on what you've learned in this course so far (and on prior knowledge), do you agree with the criticism that the design features of the most popular social media sites – such as the layout, the format (mostly text-based or mostly image-based), and algorithmic choices about what shows up first in a user's feed – make the spread of disinformation easy and likely? Why or why not?

The spread of disinformation is definitely easy and likely on the current most popular social media sites. This is due to the fact that social media algorithms push content it believes the user is more likely to have a strong reaction to, fact checked or not, keeping the user engaged on the site for as long as possible. Social media, in theory, uses the values of: creating opportunities for social connections, freedom of speech, and accessibility to entertainment to entice users to use the site. Entertainment is inherently a tricky realm as it often disguises itself in a multitude of ways including posing itself as a credible source.

Here is an example of how these values can be manipulated to encourage the spread of disinformation. Since the shareholders of a site are directly affected by its success, it is within their best interest to keep users engaged on it for as long as possible and make design choices that support their thinking. As the site becomes more personal to the user, the algorithm begins to push more and more of the content the user interacts with, creating an echo chamber of their ideas. This becomes problematic especially if the user has interacted with disinformation and may begin to build their beliefs around these ideas, potentially deciding to spread more misinformation, intentional or not. The evolving emergence of Artificial Intelligence is not helping with this either, as it opens a realm to hyper analyze the user's decisions. This may lead to potential misuse as Artificial Intelligence is even more of an unknown box than an algorithm is.

With Meta's recent decision to discontinue fact checking services, the algorithm may push more of this content to users contributing to the overall spread of disinformation. Together with the algorithm, users may be influenced and can push this content to others through the use of the share button and its counter. Regardless of the intentions of the users, they can send whatever they choose to others without any sort of warning to the validity of the statements inside of the content. This can further the spread of

disinformation and could potentially make others believe certain content is more credible depending on the amount of “shares” it has.

Although some individuals may state that fact checking and removing share counters limits a user’s freedom of speech, it is also important to consider that independent platforms have their own regulations as they are not publicly owned, and can make their own choices regarding the platforms values to the truth and the spread of disinformation. It is important to consider when developing a platform if you want to focus on controlling disinformation or allow users to post whatever they would like, even if it may be misleading to others.

X, formerly Twitter, is a great example of a privately owned business that boasts a more conservative media appearance. Elon Musk, X’s current owner, conducted many internal changes to the algorithm as well as cut the entire staff of X by over 6,000 employees. He constantly keeps a tight presence in all of X’s decision making. During the elections, for example, media feeds were overloaded with right wing media that was often promoted by the company itself.

Question 3 (15 points): Regardless of your answers to the previous question, explain what features you will implement in InstaTik in order to achieve your company's goal of "being an industry leader in the fight against the spread of disinformation". Briefly describe some of the worst-case scenarios that could result. What bad thing(s) could happen? To which stakeholders? Which stakeholders' values or interests would be harmed?

Note: you don't need to be 100% comprehensive. You only need to identify scenarios that are foreseeable, consequential, and likely. You don't need to talk about extreme events, such as alien invasions or "AI" coming to life and forcing all of us to make paper clips.

With regards to InstaTik, there will be essential checks in place in order to control for misinformation, especially for fake accounts, specifically machine learning algorithms that check for the likelihood of fake accounts, based on huge amounts of training data extracted from known fake accounts. We will also have a review team in order to ensure the plausibility of content and controlling for sensitive information, due to the drawbacks of current algorithms, since they aren't perfect. This review team would determine what is disinformation by consulting reliable news sources and correspondents on what is fact and what is fabrication, and also recognizing and flagging disinformation based on hallmark features of such content based on previously identified disinformation. More importantly, we also plan to undertake a framework to limit the ubiquity of a certain type of content being restricted to unique existence. What this means is that whenever a piece of content, such as a post or short-form video, gets uploaded to our servers, users will be asked to denote whether it is their own work or not. If not, they will cite their source of information, so as to generate a trace of origin back to the original propagator of said information contained in their piece of content.

However, if it is indeed their own work, there will be implemented an extensive review process that ensures that whatever a user has uploaded does not already exist on the platform. The motivation for such an act is that disinformation thrives off of mindless reposting and spreading on open channels, like our very own platform. As such, we will ensure that there is uniqueness in the uploading of information, and as such, we can control for misinformation and prevent duplicate uploads, which inadvertently also controls for stealing work and plagiarism. We do acknowledge the caveats, however, with such a plan. While there is a plan for how InstaTik would optimally operate, there might be issues properly implementing it in the real world. Unfortunately, there are a variety of worst-case scenarios that could result from this system. First, this approach could make it so there are 'false positive' posts. To explain, legitimate content (posted by the original

user) might be wrongly flagged as a duplicate of a repost. This could lead to misattribution, lost engagement/money, and easily stolen work. The affected stakeholders would be content creators (especially smaller ones), activists, educators, artists, etc. It could greatly affect those trying to spread a message in their own style.

Another scenario could be censorship. The algorithm might flag a legitimate expression of an issue as a duplicate if it closely matches a similar expression of the same issue. For instance, if two users or content creators want to post a 'Save The Earth' image which is expressed in different styles, one of those users could be inhibited in doing so and potentially be punished through revenue cuts or reduced platform visibility. Additionally, the line between what is misinformation and what is not can become blurred. When these rules start to get made, it can contribute to a bias of ideas in the app. This might feel like free speech is being limited, effectively silencing users. The affected stakeholders would be journalists, activists, or any other marginalized group.

There may also be a scenario where bad actors manipulate the system in place. To elaborate, users could figure out a way past the algorithm. They could alter information in order to get past duplication measures, allowing them to spread misinformation which never gets flagged. This could undermine the entire point of decreasing false information. The affected stakeholders would be the general population that consumes this content. Another scenario could be a decline in user engagement where if the algorithm is too restrictive, it could frustrate users. If they constantly can't post or see content that feels curated, those users may leave for competitors of the app. There are many people that prioritize entertainment in apps of this nature. If they can't find that in InstaTik, they would abandon it for apps that can provide it. With less users to engage with the content and advertisements of the app, advertisers and stakeholders would begin to take losses and begin to lose interest, which would contribute to the downfall of the app. The affected stakeholders would be the owners/employees of InstaTik, advertisers, investors, and shareholders.

Question 4 (20 points): It seems highly unlikely that a solely technical solution (e.g., a new set of watermarks to detect material produced by a generative model) can be developed that will resolve the debate over preventing, deterring, and slowing the spread of disinformation, in a way that satisfies all stakeholders. In the absence of such a solution, you will need to make tradeoffs between the values of different stakeholders, including your own values.

If you were able to decide the outcome, what outcome(s) would you choose, and why? In answering this, make sure to explain:

1. which stakeholders and values your outcome favors,
2. which stakeholders and values are compromised; and
3. why these value tradeoffs are appropriate.

Hint: Are any stakeholder values illegitimate in this context? Are rights that need to be respected or “red lines” that cannot be crossed? And which of the remaining values are strongest?

To begin, it’s important to establish the “red line” of values that cannot be crossed. As the developers of InstaTik, it is necessary to set our mission—combating disinformation while giving users the freedom to post what they like—as a core value that guides software and policy decisions in support of other values. If our mission’s values were compromised, then InstaTik would not be true to what it initially set out to be. Additionally, we find our mission to be ethical and self-justifiable, as it encourages the spreading of accurate and relevant information in as many ways as possible. In relation to other stakeholders, this outcome favors (besides the developers) the values of site users, content creators, and businesses.

Site users would benefit because InstaTik prioritizes access to reliable information, or at the bare minimum, significantly reduced disinformation. It also makes it much easier for the users to determine the source of the content they’re consuming. Additionally, the values of businesses would also align with the limiting of harmful disinformation, which can help protect company reputation. The accountability chain implemented by InstaTik would ensure that companies receive credit for their products if an advertisement is reposted. Furthermore, while content creation would become a more lengthy process, content creators could also benefit from the restrictions on disinformation, as a bi-product of these restrictions is that plagiarism becomes significantly more difficult. Overall, access to reliable information, the protection of reputation, and the mitigation of plagiarism are all values that are strengthened by our outcome.

However, in making this “red line”, the values of other stakeholders will inevitably be compromised. The most notable group of stakeholders at risk involve those seeking money or influence, specifically political organizations and advertisers. Recently, the spread of disinformation has become a prominent tool of political parties in the United States to either improve their own image or harm the image of their opponents. By emphasizing a mitigation of this information, it is likely that political organizations would have a more difficult time increasing their influence over the InstaTik user population. Additionally, advertisers (notably scammers) who seek to generate revenue from their product with false advertising or hyperbole would be shut down on InstaTik. In this regard, advertisers’ desire for profit would come second to the mitigation of the spread of disinformation.

I find these value tradeoffs to be acceptable, as they prioritize the company’s mission and therefore are ethically upstanding. Additionally, I find that the values of stakeholders like political organizations and advertisers wishing to generate profit (both monetary and influential) are illegitimate, as they seek selfish benefit at the cost of reliable information. From an ethical standpoint, these values are dwarfed in comparison to the values of consumers and content creators, who want reliable information and the ability to post whatever they deem fit. Making these latter values the company’s “red line” ensures that even if a stakeholder doesn’t share them, their ethical foundation justifies prioritizing them over competing interests. That said, we recognize that these values are broad, and an optimal implementation will require a broad spectrum of inputs and perspectives. In this regard, InstaTik would prioritize user feedback when adjusting how our values are manifested on the platform, thus preventing isolated development.

Question 5 (10 points): Thus far, our analysis has assumed a U.S. context. How would your proposed resolution (question 4) change if the debate were being held in a different country that does not place free speech/expression on quite the same pedestal? Why or why not? What factors are relevant here?

There are also underlying cultural differences between what really is “free speech” that persist across different nations, particularly the major divide between the individualism of the so-called “Western” world and the collectivism of the “Eastern” world. For example, content posted by elders and spiritual teachers may be regarded as sacrosanct in secular spheres of the world, whereas they may be regarded as bogus and cultist in other spheres of the world, which highlights the difference in interpretations of “free speech”.

We primarily would want to look at a cultural basis when regarding this, as often free speech and expression branch off of those aspects. When looking at platforms like TikTok or Red Note (often considered the alternative to TikTok in China), one can see a striking change. RedNote, which has almost one hundred percent Chinese users, is a much more welcoming platform compared to TikTok. That is to say, TikTok is much better compared to Instagram and X. This is categorically due to the cultural influence that China has compared to America. Respect is almost a cornerstone of Chinese culture and is often true for many other countries in Asia. TikTok interestingly has a mix of many Asian countries dominating its platform followed by a big presence in the USA.

With that aspect explained, the question of free speech/expression is not all that potent within China. Much of TikTok’s criticism came from the fact that TikTok’s parent company was owned by a Chinese company. This ultimately brings the question of reliable information. How can we state that we value reliable information if there is a threat to the integrity of that information? Reputation and plagiarism are pillars that do not get affected due to affected free speech/expression. However, is it the social media’s responsibility to provide a bypass to the lack of free speech/expression? If InstaTik was based only in China, the answer is not really. However, worldwide companies like TikTok work on different bounds. If these values are to be consistent, InstaTik would have to reside in the country that can legally provide those values (the United States).

This is not new in any way. In many ways, this is exactly what happened with TikTok. The factors that are at play here are whether this company resides inside or outside of the bound of conflict. If inside, InstaTik has no choice but to either change its values or move its place of operations elsewhere. On the outside, there is nothing that can be done unless InstaTik changes its values to heed the expectations of that country. In this case, InstaTik would most likely need to legally battle with China on the pillar of free speech/expression, as it is the company's mission. This potentially risks a ban of the InstaTik platform should no agreement or progress be made. InstaTik's proposed solution of combating disinformation while prioritizing free speech would not change in another country even if our app suffers consequences for it. It is the company's goal to be an industry leader in these fronts and changing our stance would compromise our efforts. Overall, with our app's mission, it is important to consider its availability in other countries with more restrictions around free speech.